INCH – POUND

MIL-PRF-58102A
15 May 2026
SUPERSEDING
MIL-P-58102
31 May 1978

PERFORMANCE SPECIFICATION

COVER, BREATHABLE, WATERPROOF, WEATHER RESISTANT, DURABLE, CORROSION MITIGATING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for breathable (see 6.6.3), waterproof (see 6.6.8), weather resistant (see 6.6.9), and durable (see 6.6.4) cover material intended for use to mitigate corrosion on military equipment such as weapon systems, aircraft, vehicles and trailered equipment in non-deployed applications (see 6.1).

1.2 Classification. The cover materials are furnished in the following types, classes, and grades (see 6.1):

1.2.1 Types. The types of cover material are as follows:
Type I – Aircraft cover (see 6.6.1)
Type II – Ground vehicle and equipment cover (see 6.6.6)
Type III – Shipboard systems/equipment cover (see 6.6.7)

1.2.2 Classes. The classes of cover material are as follows:
Class 1 – Standard durability
Class 2 – High durability

1.2.3 Grades. The grades of cover material are as follows:
Grade 1 – Standard
Grade 2 – Covers containing a wicking layer

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to NAVAIR-Standardization@us.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3.1 Part or Identifying Number (PIN). The part number designations for cover material acquired to this specification are created as follows:

1. The letter M
2. The general specification number
3. A dash
4. A letter designating the type:

   A – Type I
   G – Type II
   S - Type III

5. A letter designating the grade:

   S – Class 1
   H – Class 2

6. A number designating the grade:
   01 – Grade 1
   02 – Grade 2

Example: M58102 –AH03

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202-110	-	Method 110, Sand and Dust

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades Jp-5 (NATO F-44)
MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34) and NATO F-35

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MIL-DTL-53022		Primer, Epoxy Coating, Corrosion Inhibiting Lead and Chromate Free
MIL-DTL-64159	-	Camouflage Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant
MIL-PRF-23236	-	Coating Systems for Ship Structures
MIL-PRF-23377	-	Primer Coatings: Epoxy, High-Solids
MIL-PRF-24635	-	Coating Systems, Weather-Resistant, Exterior Use
MIL-PRF-85285	-	Topcoat, Aircraft and Support Equipment

(Copies of these documents are available online at https://quicksearch.dla.mil.)

CODE OF FEDERAL REGULATIONS

Code of Federal Regulations, Title 40	-	Protection of Environment

(Copies of this document are available online at https://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILES CHEMISTS AND COLORISTS (AATCC)

TM118	-	Test Method for Oil Repellency: Hydrocarbon Resistance
TM183	-	Transmittance or Blocking of UV Radiation Through Fabric

(Copies of these documents are available online at https://www.aatcc.org/.)

ASSOCIATION OF MATERIALS PROTECTION AND PERFORMANCE (AMPP)

SSPC-PS 10/ NACE	-	Near-White Metal Blast Cleaning No. 2

(Copies of this document are available online at https://www.ampp.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/ASHRAE

ANSI/ASHRE Standard 52.2	-	Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size
ANSI/ESD STM11.13	-	The Protection of Electrostatic Discharge Susceptible Items Two-Point Resistance Measurement

(Copies of these documents are available online at https://webstore.ansi.org/.)

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ASTM INTERNATIONAL

ASTM A1008/A1008M	-	Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM B117	-	Standard Method of Salt Spray (Fog) Testing
ASTM D523	-	Standard Test Method for Specular Gloss
ASTM D751	-	Standard Test Methods for Coated Fabrics
ASTM D2261	-	Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure (Constant-Rate-of-Extension Tensile Testing Machine)
ASTM D2843	-	Standard Test Method for Density of Smoke from the Burning or Decomposition of Plastics
ASTM D3389	-	Standard Test Method for Coated Fabrics Abrasion Resistance (Rotary Platform Abrader)
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3786/D3786M	-	Standard Test Method for Bursting Strength of Textile Fabrics-Diaphragm Bursting Strength Tester Method
ASTM D4417	-	Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel
ASTM D4833/D4833M	-	Standard Test Method for Index Puncture Resistance of Geomembranes and Related Products
ASTM D4966	-	Standard Test Method for Abrasion Resistance of Textile Fabrics (Martindale Abrasion Tester Method)
ASTM D5035	-	Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method)
ASTM D6413/D6413M		Test Method for Flame Resistance of Textiles (Vertical Test)
ASTM E662	-	Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials
ASTM E96/E96M	-	Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM G154	-	Standard Practice for Operating Fluorescent Ultraviolet (UV) Lamp Apparatus for Exposure of Materials

(Copies of these documents are available online at https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 9237	-	Determination of the Permeability of Fabrics to Air

(Copies of this document are available online at https://www.iso.org.)

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SAE INTERNATIONAL

SAE J1966 - Lubricating Oils, Aircraft Piston Engine (Non-Dispersant)

(Copies of this document are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) samples shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Corrosion mitigating covers shall be made from such materials and processes as to ensure compliance with the requirements of this specification.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Performance requirements. The type, class, and grade of cover material, as specified in the contract (see 6.2), shall conform to the applicable requirements specified in Table I when tested in accordance with 4.4.

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TABLE I. Performance requirements.

Characteristics	Type I Requirements	Type II Requirements	Type III Requirements	Test Paragraph Reference
Dry weight oz./yd^2 (grams/m^2) [Grade] (max)	7(240) [all grades]	25(857) [1,3] 35(1200) [2]	25(857) [1,3] 35(1200) [2]	4.4.1.1
Wet weight (avg. % wt. gain) [Grade] (max)	10 [all grades]	155	155 [1,3]	4.4.1.2
Drying rate (avg hours) [Grade]	1 [all grades]	72 [1,3] 80 [2]	72 [1,3] 80 [2]	4.4.1.3
Breaking strength, avg Lb-force (Newtons) (min) a. Class 1 Warp b. Class 1 Fill c. Class 2 Warp d. Class 2 Fill	150(667) >50% of Warp 350(1556) >50% of Warp	150(667) >50% of Warp 350(1556) >50% of Warp	150(667) >50% of Warp 350(1556) >50% of Warp	4.4.2.1
Bursting strength, avg PSI (Kilopascals) (min) (when specified (see 6.2)) a. Class 1 b. Class2	250 (1724) 400 (2758)	250 (1724) 400 (2758)	250 (1724) 400 (2758)	4.4.2.2
Tearing strength, avg Lb-force (Newtons) (min) a. Class 1 Warp b. Class 1 Fill c. Class 2 Warp d. Class 2 Fill	20(89) >50% of Warp 80(356) >50% of Warp	20(89) >50% of Warp 80(356) >50% of Warp	20(89) >50% of Warp 80(356) >50% of Warp	4.4.2.3
% Elongation, avg (max) a. Class 1 Warp b. Class 1 Fill c. Class 2 Warp d. Class 2 Fill	10% 10% 5% 5%	10% 10% 5% 5%	10% 10% 5% 5%	4.4.2.4
Abrasion resistance, avg Cycles (min) a. Class 1 Warp b. Class 1 Fill c. Class 2 Warp d. Class 2 Fill	1500 1500 9000 9000	1500 1500 9000 9000	1500 1500 9000 9000	4.4.2.5
Puncture Resistance, Lb-force (Newtons) **a.** Screwdriver **b.** Rod	75 (333.6) 250 (1112)	75 (333.6) 250 (1112)	75 (333.6) 250 (1112)	4.4.2.6
Hydrostatic resistance	<2 droplets	<2 droplets	<2 droplets	4.4.3.1

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Table I. Performance requirements – Continued.

Characteristics	Type I Requirements	Type II Requirements	Type III Requirements	Test Paragraph Reference
MVTR (min)	0.03 oz./yd^2/hour (1.0 grams/m^2/hour)	0.03 oz./yd^2/hour (1.0 grams/m^2/hour)	0.03 oz./yd^2/hour (1.0 grams/m^2/hour)	4.4.3.2
Resistance to accelerated aging (avg from original) (min)	50% breaking strength 20% tearing strength 90% abrasion resistance	50% breaking strength 50% tearing strength 90% abrasion resistance	50% breaking strength 50% tearing strength 90% abrasion resistance	4.4.3.3
UV transmission	<5%	<5%	<5%	4.4.3.4
Flex-fatigue in wind (when specified (see 6.2))	No holes, tears, or delamination	No holes, tears, or delamination	No holes, tears, or delamination	4.4.3.5
Color	According to contract requirements	According to contract requirements	According to contract requirements	4.4.4.1
Color fade after accelerated weathering	No change in color	No change in color	No change in color	4.4.4.2
Glossiness 1/	at 60° ≤ 2 units at 85° ≤ 2 units	at 60°≤ 2 units at 85°≤ 2 units	at 60°≤ 2 units at 85°≤ 2 units	4.4.4.4
Flame resistance	N/A	N/A	Char length < 3.5 in (8.9 centimeters) 10 seconds after flame No flaming droplets or debris	4.4.5.1
Flame retardancy	N/A	N/A	Flame shall not pass top edge of specimen during 42 second interval; no dripping	4.4.5.2
Smoke opacity (max) 2/	N/A	N/A	450	4.4.5.3
Smoke composition (max) 2/	N/A	N/A	50	4.4.5.4
Blocking	Blocking rating ≤2	Blocking rating ≤2	Blocking rating ≤2	4.4.6.1
Sand resistance (when specified (see 6.2))	No scratches or abrasions	N/A	N/A	4.4.6.2
Cover material abrasiveness on canopy/window materials (when specified (see 6.2))	No scratches or abrasions	N/A	N/A	4.4.6.3

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Table I. Performance requirements – Continued.

Characteristics	Type I Requirements	Type II Requirements	Type III Requirements	Test Paragraph Reference
Cover material abrasiveness on paint materials	No scratches or abrasions	N/A	N/A	4.4.6.4
Surface resistance [Grade] 3/	1 x 10^5 Ω ≤ surface resistance < 1 x 10^{11} Ω [3]	N/A	N/A	4.4.6.5
Dust penetration 4/ (when specified (see 6.2))	99.5% rejection of particles ≥1.0 microns	N/A	N/A	4.4.6.6
Resistance to oil (avg) (when specified (see 6.2))	oil holdout rating ≥6	N/A	N/A	4.4.6.7
Resistance to aromatic hydrocarbon fluid	No discoloration, swelling, stretching, or loss of dimension	No discoloration, swelling, stretching, or loss of dimension	No discoloration, swelling, stretching, or loss of dimension	4.4.6.8
Adherence to coated substrate	No adherence or material transfer	No adherence or material transfer	No adherence or material transfer	4.4.6.9
Breathability (min)	150 mm/s	150 mm/s	150 mm/s	4.4.6.10
Wicking rate (min) [Grade]	<5 mins to 20mm <30 mins to 150mm	<5 mins to 20mm <30 mins to 150mm	<5 mins to 20mm <30 mins to 150mm	4.4.6.11

1/ There shall be low gloss on the outside face of the cover material or both sides if useable with either side to the weather.
2/ The cover material shall have no adverse health effects in both its burning and non-burning states.
3/ This shall be the surface resistance on both the front and back of the cover material.
4/ The cover material shall filter out sand, dust, and pollutants to the specified level of rejection.

3.4 Resistance to accelerated aging. When testing in accordance with 4.4.3.3, the average breaking strength following UV exposure shall be at least 50 percent of the baseline breaking strength measured (see Table I and 4.4.2.1.). The average tearing strength following UV exposure shall be at least 20 percent of the baseline tearing strength for Type I cover material and 50 percent of the baseline tearing strength for Type II cover material (see Table I and 4.4.2.3.). The average abrasion resistance following UV exposure shall be at least 90 percent of the baseline abrasion resistance (see Table I and 4.4.2.5.)

3.5 Back side identification. If the cover material has only one side intended to be toward weather when used, the back side shall be identified by applying a stamping or other suitable means onto the cover material with, as a minimum, the word "Back" at the beginning and end of each roll. The font shall be at least 2 inches wide by 1 inch height. It shall be a reoccurring pattern with a horizontal spacing of 6 inches between each instance of the word, “Back.” The labeling shall not show through onto the face side of the cover material. The cover material shall be visually examined and meet the requirements of 4.4.4.3.

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3.6 Form. The cover length and width or custom pattern shall be as specified in the contract (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified (see 6.2), first article inspection shall consist of all examinations and tests listed in this specification for the applicable type, class, and grade of cover material.

4.3 Conformance inspection. Conformance inspection shall consist of the tests specified in Table II. Due to custom sizes and typically low quantities, sampling for inspection shall be as specified in the contract (see 6.2).

TABLE II. Conformance (C) testing requirements.

Test	Test Para.	All Types
Dry weight	4.4.1.1	X
Breaking strength (W/F)	4.4.2.1	X
Bursting strength	4.4.2.2	X
Tearing strength (W/F)	4.4.2.3	X
Hydrostatic resistance	4.4.3.1	X
MVTR	4.4.3.2	X
Color	4.4.4.1	X
Blocking	4.4.6.1	X

4.4 Test methods. Unless otherwise specified, all testing shall be conducted in accordance with the procedures of 4.4.

4.4.1 Weight.

4.4.1.1 Dry weight. Five 6-inch by 6-inch (15-centimeter by 15-centimeter) samples shall be tested. The weight shall be determined in accordance with ASTM D3776/D3776M Option B at 20 percent relative humidity (RH). The average weight shall meet the requirements specified in Table I.

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4.4.1.2 Wet weight. The samples tested in 4.4.1.1 shall be immersed in water for two hours and allowed to drip dry for five minutes. The weight shall be determined in accordance with ASTM D3776/D3776M Option B at 20 percent relative humidity (RH). The percent weight gain shall be determined as follows:

$$Percent\ Weight\ Gain = \frac{Wet\ Weight - Dry\ Weight}{Dry\ Weight} x100$$

The average percent weight gain shall meet the requirements specified in Table I.

4.4.1.3 Drying rate. The samples tested in 4.4.1.2. shall be placed in an environmental chamber at 20 percent RH and 70 °F and hung in a vertical orientation to dry. The weight shall be determined in accordance with ASTM D3776/D3776M Option B every ten minutes for the first two hours and then at twelve-hour intervals thereafter. The balance shall be wiped dry in between each measurement. The total time to reach a weight of 105 percent (or less) of the original dry weight established in 4.4.1.1 will be recorded as the dry time. The average dry time shall meet the requirements specified in Table I.

4.4.2 Physical durability.

4.4.2.1 Breaking strength. Ten 1-inch by 6-inch (2.5-centimeter by 15.2-centimeter) samples shall be tested, five in the warp direction and five in the fill direction, and the breaking strength shall be determined in accordance with ASTM D5035 1C-E or 1C-T. The average breaking strength in the warp and fill directions shall meet the requirements specified in Table I.

4.4.2.2 Bursting strength. Ten 5-inch by 5-inch (12.7-centimeter by 12.7-centimeter) samples shall be tested. The bursting strength shall be determined in accordance with ASTM D3786/D3786M. The average bursting strength shall comply with Table I.

4.4.2.3 Tearing strength. Ten 3-inch by 8-inch (7.5-centimeter by 20-centimeter) samples shall be tested, five in the warp direction and five in the fill direction. The tearing strength shall be determined in accordance with ASTM D2261. The average tearing strengths in the warp and fill directions shall meet the requirements specified in Table I.

4.4.2.4 Elongation. Ten 1-inch by 6-inch (2.5-centimeter by 15.2-centimeter) samples shall be tested, five in the warp direction and five in the fill direction. The elongation shall be determined in accordance with ASTM D5035 1C-E or 1C-T. The average maximum elongation in the warp and fill directions shall meet the requirements in Table I.

4.4.2.5 Abrasion resistance. Five circular samples 4.5 inches (11 centimeters) in diameter shall be tested, and the number of cycles to full penetration shall be determined in accordance with ASTM D3389. The rotary abrasion tester shall have double heads with a 17.6-ounce (500-gram) auxiliary weight and H-18 wheels. The test shall be continued until either a full penetration to the back plate is achieved or to 10,000 cycles, whichever first occurs. The average abrasion resistance shall meet the requirements in Table I.

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4.4.2.6 Puncture resistance. The puncture resistance using a screwdriver probe shall be determined in accordance with ASTM D751. The puncture resistance using a rod probe shall be determined in accordance with ASTM D4833/D4833M. The samples shall meet the requirements of Table I.

4.4.3 Weatherability.

4.4.3.1 Hydrostatic resistance. The hydrostatic resistance shall be determined in accordance with ASTM D751, Hydrostatic Resistance Procedure A, Procedure 2 at 1 atm. After five minutes test duration, inspect the test specimen. The test specimen shall meet the requirements in Table I.

4.4.3.2 Moisture vapor transmission rate (MVTR). The moisture vapor transmission rate shall be determined in accordance with ASTM E96/E96M, Procedure for Water Method, and shall meet the requirements in Table I.

4.4.3.3 Resistance to accelerated aging. Twenty-five (25) samples shall be tested. Ten samples shall be as specified in 4.4.2.1, ten samples shall be as specified in 4.4.2.3, and five samples shall be as specified in 4.4.2.5. Color shall then be tested using samples specified in 4.4.2.3 and tested in accordance with 4.4.4.1. These same 4.4.2.3 color determination samples shall then be used to test the gloss in accordance with 4.4.4.4. All samples shall then be exposed for 1000 hours UV in accordance with ASTM G154, Cycle 1 of Table X2.1. Following exposure, the gloss requirement samples shall be tested in accordance with 4.4.4.4 and color fade shall be tested in accordance with 4.4.4.2. The average breaking strength, tearing strength, and abrasion resistance of exposed samples shall then be determined, in order, by testing five samples in accordance with 4.4.2.1, 4.4.2.3, and 4.4.2.5, respectively. Samples shall meet the requirements of Table 1.

4.4.3.4 UV transmission. Three 3-inch by 3-inch (7.5-centimeter by 7.5 centimeter) samples shall be tested. The transmittance of UVA radiation through the cover sample shall be determined in accordance with AATCC TM183 and shall meet the requirements in Table I.

4.4.3.5 Flex-fatigue in wind. When specified (see 6.2), construct three flag samples out of the cover material complying with figure 3. The 12 mm channel shall be created by stitching the material according to the manufacturer's recommended method. The flag shall be installed in the wind tunnel by threading a length of 3/8 in diameter rod through the 0.5 in sleeve. The rod shall be fitted with a clamping collar of 7/8 in minimum outer diameter on either side of the flag, as shown in figure 4. The rod and flag assembly shall be affixed in the observation chamber of the wind tunnel such that the axis of the tube is parallel with the benchtop. Expose each flag to 100 mile per hour (mph) winds in an open loop benchtop wind tunnel for two hours. Following testing, visually inspect each sample for compliance with Table I.

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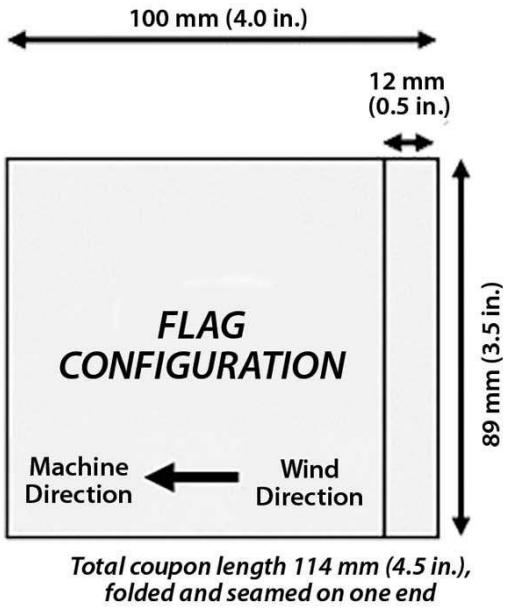


FIGURE 3: Flag sample configuration for flex fatigue in wind testing.

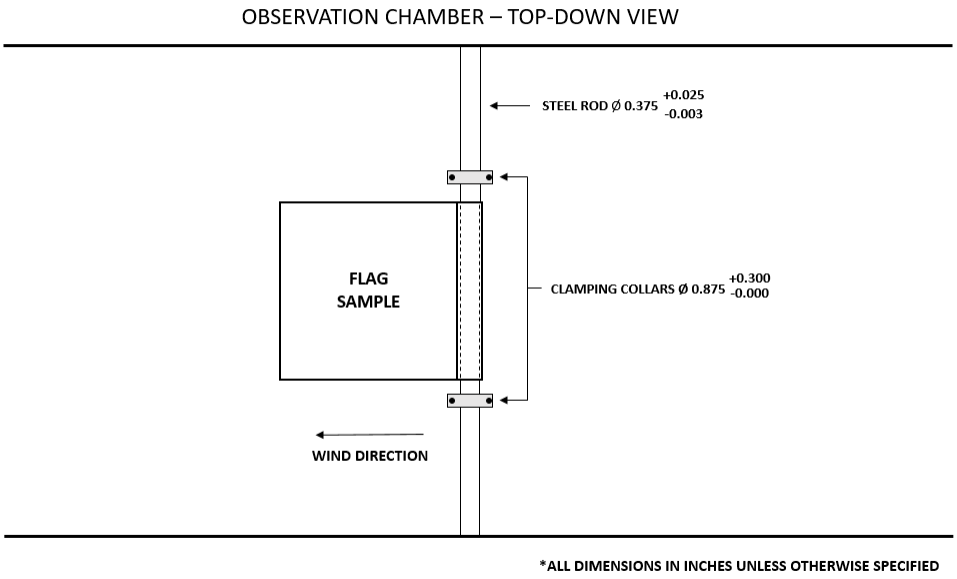


FIGURE 4. Flex fatigue in wind test chamber setup.

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4.4.4 Visual and spectral testing procedures.

4.4.4.1 Color. The samples tested as (see 6.11.1 and 6.2c) specified by 4.4.2.3 in 4.4.3.3 will be used for color determination. Samples shall be visually inspected to meet the requirements in Table I.

4.4.4.2 Color fade after accelerated weathering. Three samples used in 4.4.4.1 shall be exposed for 1000 hours UV exposure in accordance with Cycle 1 of ASTM G154 during 4.4.3.3. Following exposure, the color of the exposed samples shall be visually compared to the unexposed samples for compliance with Table I.

4.4.4.3 Back side identification. A sample of the cover material shall be tested. The cover shall be laid on an opaque surface such as the ground or a table. Upon visual inspection, it shall pass according to the requirements of 3.5.

4.4.4.4 Glossiness. Unless otherwise specified, gloss measurements shall meet the requirements specified in Table I when tested in accordance with ASTM D523. Test samples shall be as specified in 4.4.2.3 and tested at 60° and 85° geometries. When specified (see 6.2), glossiness other than the Table I value shall be ≤ 8 gloss units.

4.4.5 Fire characteristics (Type III only).

4.4.5.1 Flame resistance. The flame resistance shall be determined in accordance with ASTM D6413/D6413M and shall meet the requirements specified in Table I.

4.4.5.2 Flame retardancy. Five test samples shall be tested. The flame retardancy shall be determined in accordance with ASTM D6413/D6413M except that the specimen shall be observed for an additional 30 seconds after burner flame turn off. If the tip of the flame passes beyond the top edge of the specimen at any time during the 42 seconds after the burner is turned on, the specimen shall be reported as a test failure. Five specimens shall be tested in the warp direction and the results reported individually as “pass” or “fail.” The test results shall meet the requirements specified in Table I.

4.4.5.3 Smoke opacity. Smoke opacity testing shall be conducted in accordance with ASTM E662 and shall meet the requirements specified in Table I.

4.4.5.4 Smoke composition. Smoke composition testing shall be conducted in accordance with ASTM D2843 and shall meet the requirements specified in Table I.

4.4.5.5 Health/hazard assessment. When specified by the procuring activity (see 6.2), the cover material shall be evaluated by the Navy and Marine Corps Force Health Protection Command (NMCFHPC), using the administrative HHA (see 6.4).

4.4.6 Other testing procedures.

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4.4.6.1 Blocking. Three samples shall be tested. The blocking resistance of the cover material shall be determined in accordance with ASTM D751, except that the specimens shall be exposed at an oven temperature of 180 ± 2 °F (82 ± 1 °C) for 30 minutes. The average rating shall meet the requirements specified in Table I.

4.4.6.2 Sand resistance. When specified (see 6.2), three circular cover samples 1.5 inches (3.8 centimeters) in diameter shall be acquired. Samples shall be immersed in sand that is in accordance with the dust requirements specified in MIL-STD-202-110. The samples shall then be hung vertically to allow loose sand to fall off the sample. The samples shall be placed in a Martindale abrasion tester as described in ASTM D4966. Three circular discs 1.5 inches (3.8 centimeters) in diameter constructed of clear acrylic shall be used for testing on each testing table as opposed to the standard fabric described in ASTM D4966. Run the abrasion tester for 10,000 rubs at 1.74 psi (12 kPa) in accordance with ASTM D4966. After testing visually inspect the acrylic discs and samples for compliance with the requirements of Table I.

4.4.6.3 Cover material abrasiveness on canopy and window materials. When specified (see 6.2), three circular cover samples 1.5 inches (3.8 centimeters) in diameter shall be acquired. The samples shall be placed in a Martindale abrasion tester as described in ASTM D4966. Three circular discs 1.5 inches (3.8 centimeters) in diameter constructed of clear acrylic shall be used for testing on each testing table as opposed to the standard fabric described in ASTM D4966. Run the abrasion tester for 10,000 rubs at 1.74 psi (12 kPa) in accordance with ASTM D4966. After testing visually inspect the acrylic discs for compliance with the requirements of Table I.

4.4.6.4 Cover material abrasiveness on paint materials (Type I only). Six circular cover samples 1.5 inches (3.8 centimeters) in diameter shall be acquired. The samples shall be placed in a Martindale abrasion tester as described in ASTM D4966. Six circular discs 5.5 inches (14 centimeters) in diameter constructed of ASTM A1008/A1008M steel shall be acquired. Panels shall be prepared by abrasively blasting to SSPC-SP 10/NACE No. 2 with a 2-4 mil profile as determined by ASTM D4417. Three of the steel discs shall be coated with MIL-PRF-23377 Type I or II Class N primer and then top coated with MIL-PRF-85285 Type I or III in accordance with the appropriate manufacturer product data sheets. Three of the steel discs shall be coated with MIL-DTL-53022 IV primer and then top coated with MIL-DTL-64159 (CARC coating) Type II in accordance with the manufacturer product data sheet. The coated steel discs shall be used for testing on each testing table as opposed to the standard fabric described in ASTM D4966. Run the abrasion tester for 10,000 rubs at 1.74 psi (12 kPa) in accordance with ASTM D4966. After testing visually inspect the coated steel discs for compliance with Table I.

4.4.6.5 Surface resistance. The surface resistance shall be determined in accordance with ANSI/ESD STM 11.13 and shall meet the requirements of Table I.

4.4.6.6 Dust penetration. When specified (see 6.2), three cover samples shall be tested. Determine the percent rejection of particles 1.0 microns or larger in accordance with ANSI/ASHRE Standard 52.2. The percent rejection shall meet the requirements specified in Table I.

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4.4.6.7 Resistance to oil. When specified (see 6.2), three cover samples shall be tested. Determine the oil holdout rating in accordance with AATCC 118 using oil in accordance with SAE viscosity grade 50 of SAE J1966. The average holdout rating shall comply with Table I.

4.4.6.8 Resistance to aromatic hydrocarbon fluid. Six 6-inch by 6-inch (15-centimeter by 15-centimeter) samples shall be acquired. Immerse three of the samples in JP-5 fuel that conforms to MIL-DTL-5624 and store for 24 hours. Immerse the remaining three samples in JP-8 fuel that conforms to MIL-DTL-83133 and store for 24 hours. Remove the samples and hang in a vertical position for 24 hours. Inspect the cover material for discoloration, swelling, stretching, or shrinkage and confirm the samples meet the requirements specified in Table I.

4.4.6.9 Adherence to coated substrate. Three aluminum panels approximately 12-inch by 12-inch by ¼-inch (300 by 300 by 6 millimeters) in size shall be used for testing. Panels shall be prepared by abrasively blasting to SSPC-SP 10/NACE No. 2 with a 2-4 mil profile in accordance with ASTM D4417. One panel shall be coated with MIL-DTL-53022 IV primer in accordance with the manufacturer product sheet. The panel shall then be top coated with MIL-DTL-64159 (CARC coating) Type II in accordance with the manufacturer product data sheet. The second panel shall be coated with MIL-PRF-23236 Type V, VI, or VII primer in accordance with the manufacturer product data sheet. The panel shall then be top coated with MIL-PRF-24635 Type II, III, or IV in accordance with the manufacturer product data sheet. The third panel shall be coated with MIL-PRF-23377 Type I or II Class N primer in accordance with the manufacturer product data sheet. The panel shall then be top coated with MIL-PRF-85285 Type I or III in accordance with the manufacturer product data sheet.

Three test cover materials samples shall be cut to fully cover each of the three coated test panels on one side. Before covering each panel with its test material, spray water (tap water) to wet the entire surface. Each panel shall then be covered with test material followed by one layer of plastic wrap. Four aluminum panels ¼ inch thick, 3-inch square shall be applied at least 1 inch apart and at least 1 inch from edge of larger panel. Each of these shall be weighted to obtain 10 pounds total force. The panels shall remain in place for 1 week, indoors, at room temperature. The ambient temperature and humidity shall be recorded.

After 1 week, carefully remove the weighted samples and plastic wrap, leaving the cover material in place. Then carefully remove the cover material and observe the four compressed areas on the underside of the cover samples, as well as the contact points on the coated surface. Record any occurrence of adherence between the cover material and coating, for instance, moderate amount of coating adhering to the cover material or small amount of cover material adhering to the coating. Each panel shall be given a pass or fail rating in accordance with Table I.

4.4.6.10 Breathability. Three samples shall be tested. Samples shall be prepared and tested as specified in ISO 9237. Samples shall meet the requirements of Table I.

4.4.6.11 Wicking. Three samples shall be tested. Samples shall be prepared and tested as specified in AATCC TM 197. Wicking rate shall be measured on the side of the sample that will serve as the bottom surface of the cover that comes in contact with the asset being protected. Samples shall meet the requirements listed in Table I.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the requirements for breathable, waterproof, weather resistant, and durable cover material intended for use in non-deployed applications as protective covers for military equipment (e.g., weapon systems, aircraft, vehicles, and trailered equipment such as generators, water tanks, portable heating and cooling units, gas turbine air starters, light sets, etc.). The principal difference in this cover material over traditional material is the combination of waterproofness and breathability. This allows the material to repel dirt, dust and liquid water while allowing air and water vapor to pass through, thus reducing buildup of condensed moisture under the covering. This has the potential for significant reduction of corrosion and related degradation on the item covered. Two levels of durability are defined within the specification to meet end-user requirements.

It is not intended that all the requirements or tests contained herein will be invoked for any particular cover. Acquisition requirements should be tailored to and should contain only appropriate requirements from this specification along with any other requirements deemed necessary by the procuring activity for the specific application.

6.1.1 Color. The cover material should be available in black, desert tan, forest green, and haze gray. For color selection, SAE-AMS-STD-595 should be used.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type, Class, and Grade required (see 1.2)
c. Color and standard sample required (see Table I and 6.1.1)
d. Length and width or custom pattern configuration required (see 3.6)
e. First Article Inspection, if required (see 3.1)
f. Sampling plan for conformance, if required (see 4.3)
g. Flex fatigue in wind resistance, if required (see 4.4.3.5)
h. Glossiness rating, if other than Table I value (see 4.4.4.4)
i. Health Hazard Assessment, if required (see 4.4.5.5 and 6.4)
j. Sand resistance, if required (see 4.4.6.2)

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k. Resistance to abrade canopy/window material, if required (see 4.4.6.3)
l. Dust penetration resistance, if required (see 4.4.6.6)
m. Resistance to oil, if required (see 4.4.6.7)
n. Packaging requirements (see 5.1)
o. Joining or sewing procedures recommended by the cover material supplier
p. Cover material repair procedures recommended by the supplier

6.3 Requirements. Requirements were established to the extent possible based on previous DoD research projects. See reports NRL 6130/1567 “Test and Evaluation of Corrosion-Mitigating Covers” and VPS report No. SEA3-1505.01 “CPPMP Evaluation of Vehicle and Equipment Covers”.

6.4 Health/hazard assessment (HHA). The procuring activity will request the Navy and Marine Corps Force Health Protection Command (NMCFHPC) to perform an HHA on the barrier material furnished. The results of the NMCFHPC HHA must be deemed acceptable by the procuring activity. A flowchart for the HHA process can be found as enclosure (1) of BUMEDINST 6270.8. Additionally, information can be found on the website: https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquistion-Technical-Support/Health-Hazard-Assessments-HHA/. The HHA is a review of the product based on information submitted by the manufacturer, to assess health hazards associated with the handling, application, use and removal of the product. In order to process the HHA, NMCFHPC will request manufacturer to provide sufficient information be sent to Commander, the Navy and Marine Corps Force Health Protection Command, Industrial Hygiene – HHA, 620 John Paul Jones Circle, Suite 1100, Portsmouth, VA 20378-2103 (see 6.2).

6.5 Safety Information. Contracting officers will identify those activities requiring copies of the HHA and/or TC be provided to activities utilizing the material in order to relay appropriate health/safety, industrial hygiene, and/or material disposal recommendations.

6.6 Definitions.

6.6.1 Aircraft covers. A cover used to protect an aircraft exterior from weather/sunlight exposure. Such covers are typically intended to fit snugly on the aircraft to allow smooth runoff of precipitation. Aircraft covers aim to resist damage from wind whipping and abrasion of both the cover material and aircraft surfaces (see 1.1).

6.6.2 Blocking. The tendency of a material to stick to itself when stacked or folded (see 4.4.6.1).

6.6.3 Breathable. Able to allow air and water vapor to pass through as specified herein as moisture vapor transmission rate (see 1.1).

6.6.4 Durable. Able to withstand handling including folding/unfolding, installing/uninstalling, occasional storage when wet, washing (if allowed), etc. (see 1.1).

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6.6.5 ESD sensitive equipment covers. Covers providing weather and sunlight protection to items that include equipment sensitive to electrostatic discharge such as electronic sensors. (see 1.2).

6.6.6 Ground vehicle and equipment covers. Covers for weather, sunlight, and corrosion protection; intended for exterior equipment stored on land in a non-marine environment. (see 1.2).

6.6.7 Shipboard covers. Covers for weather, sunlight, and corrosion protection; intended for exterior shipboard deck mounted equipment. (see 1.2).

6.6.8 Waterproof. Able to block the passage of liquid water through a material as specified herein as hydrostatic resistance (see 1.1).

6.6.9 Weather resistant. Able to withstand exposure to natural weather stresses including wind, precipitation, dirt, dust, sunlight exposure, heat, and cold (see 1.1).

6.7 Subject term (key word) listing.

Aluminum
Exterior
Steel

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - SM
Navy - AS
Air Force - 69

Preparing activity:
Navy - AS

(Project 8135-2025-002)

Review activities:
Army – AT, AV
Navy - SH
Air Force - 110
DLA - DH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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